

Fig. 5: Actual-size, single-side PCB for deriving 5V from the portable wind turbine

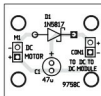


Fig. 6: Component layout of the PCB

module (board 1) through polarity guard diode 1N5817 (D1), because the motor rotation is in counter-clockwise direction. The PFM module is connected across buffer capacitor (C1) to get a USB-standard 5V DC output through its USB (A-type) connector.

## Construction and testing

An actual-size, single-side PCB for deriving 5V from portable wind turbine is shown in Fig. 5 and its component layout in Fig. 6.

Good air flow is essential for the proper working of the circuit. You

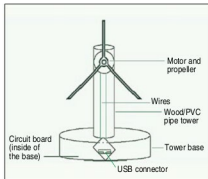


Fig. 7: Proposed setup of the portable wind turbine

can use an ordinary electric fan (or table fan) or an electric air blower as the wind-energy source for testing purpose. Ensure that air flow from the wind source directly falls at the front of the propeller. The propeller should rotate at a fast speed (in counter-clockwise direction) to get higher output.

You need rigid mounting for the wind turbine system. For this, you can fix the motor on a suitable-sized

PVC/metal pipe and attach it to a base support as shown in Fig. 7. The base support can be a small-diameter disc made of wood or metal. Connect the PFM module to the circuit at CON1, enclose these in a suitable cabinet and place inside the base in such a way that you can access the USB port conveniently.

Next, route the wires from the DC motor through the pipe to the base and connect the wires to the circuit at M1. Face the propeller towards a wind source. Now, you are ready to get 5V DC output from the USB connector. This output can also be used for charging your mobile phone.

**Caution!** Rotating blades are sharp and can cause severe injury if not handled properly. ●



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